



AI Spending Habit Analyzer: An Intelligent Expense Monitoring and Overspending Prediction System

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PROJECT OVERVIEW

The AI Spending Habit Analyzer is an intelligent expense monitoring system that uses Artificial Intelligence and Machine Learning to analyze user spending behavior and predict potential overspending risks. The system allows users to input or record their daily expenses, categorize transactions, and view financial summaries. Using historical spending data, the system identifies patterns and generates predictive insights to help users improve financial decision-making.

PROJECT STATEMENT

Many individuals fail to manage their finances effectively due to a lack of awareness regarding their spending habits. Overspending often occurs without early warning signs, leading to financial stress, debt accumulation, and difficulty achieving savings goals. Existing expense tracking applications mainly focus on recording transactions and generating historical reports rather than predicting future spending behavior. As a result, users may only realize financial problems after they have already occurred. There is a need for an intelligent expense monitoring system capable of analyzing spending patterns and predicting potential overspending risks before they happen. The proposed AI Spending Habit Analyzer addresses this problem by utilizing machine learning algorithms to evaluate spending behavior and generate predictive financial insights.

TARGET USERS AND THEIR NEEDS

Target users	Needs
Students	Monitor daily expenses and avoid overspending within a limited allowance.
Young Professionals	Manage monthly income and improve budgeting habits.
Families	Track household expenses and maintain a balanced budget.
Individuals	Analyze spending patterns and receive personalized financial recommendations.

KEY REQUIREMENTS

FUNCTIONAL REQUIREMENTS

- Allow users to register and log into the system.
- Record, edit, and delete expense transactions.
- Categorize expenses.
- Store transaction records in a database.
- Analyze spending history using a Random Forest model.
- Predict overspending risks.
- Generate personalized financial recommendations.
- Display expense summaries and spending trends through a dashboard.

NON-FUNCTIONAL REQUIREMENT

- Provide a responsive and user-friendly web interface.
- Securely store user financial data.
- Generate predictions with minimal delay.
- Maintain reliable prediction performance using standard evaluation metrics.
- Support future system enhancements and larger datasets.

DETAILED USE CASES

Use Case Narrative 1

Field	Description
Use Name Case	Record Expense
Primary Actor	User
Description	Allows the user to record an expense by entering the amount, category, date, and optional

	notes. The system validates the input and stores the transaction in the database.
Preconditions	System is running and user is logged in
Main Flow	1. User opens the Add Expense page. 2. User enters the expense details. 3. User submits the transaction. 4. The system validates the input. 5. The system saves the expense record. 6. The dashboard updates the expense summary.
Postconditions	The expense is successfully stored and becomes available for reports and analysis.
Alternative Flow	If required fields are missing or invalid, the system displays an error message and requests valid input.

Use Case Narrative 2

Field	Description
Use Name Case	Predict Overspending Risk
Primary Actor	User
Description	System analyzes spending and predicts overspending.
Preconditions	The user is logged in and has sufficient expense records.
Main Flow	1. User requests a spending analysis. 2. The system retrieves the user's expense history. 3. The system preprocesses the data. 4. The Random Forest model generates a prediction.

	5. The system displays the predicted risk level and financial recommendations.
Postvonditions	The prediction results and recommendations are displayed on the dashboard.
Alternative Flow	Insufficient data prompts user to add more records before a prediction can be generated,

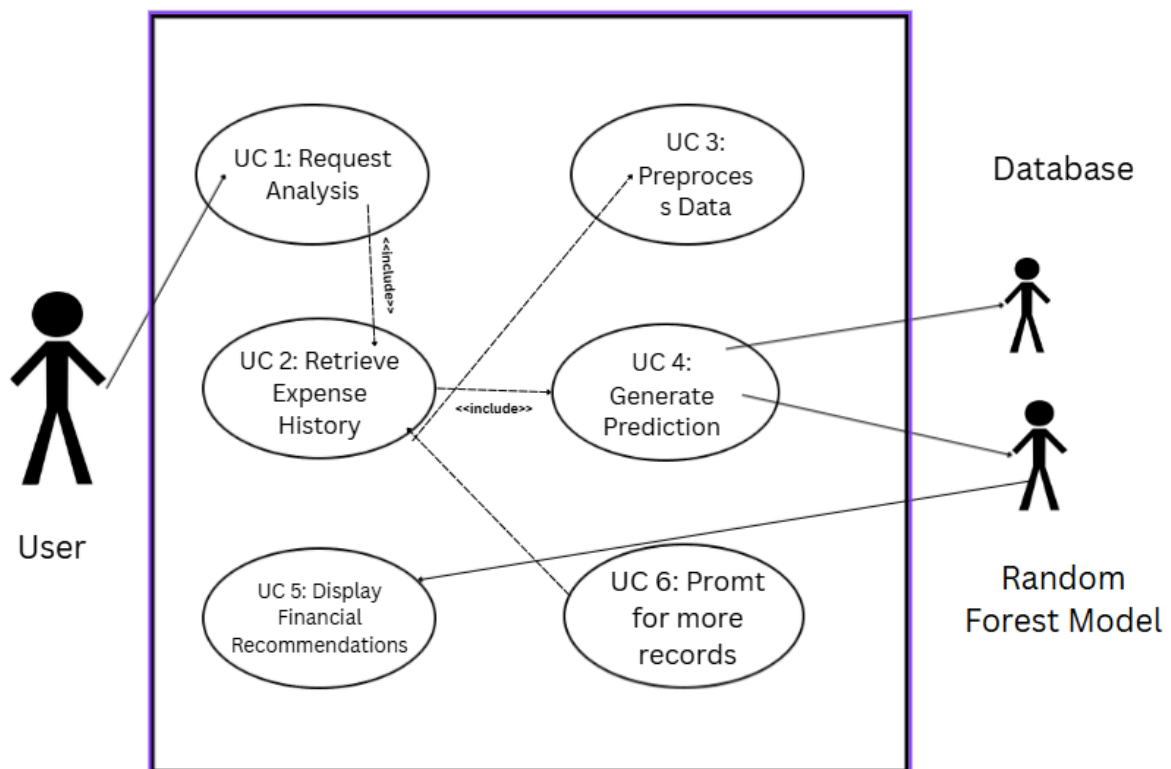


Figure 1: Use Case Diagram of the AI Spending Habit Analyzer.

OBJECTIVES

GENERAL OBJECTIVE

To develop an AI-powered expense monitoring system capable of analyzing spending habits and predicting potential overspending behavior.

Specific Objectives

1. To collect and prepare expense data for machine learning analysis.

- 2. To develop a Random Forest model that predicts overspending risks based on users' spending behavior.
- 3. To design a web-based system for recording, managing, and visualizing expense transactions.
- 4. To provide personalized financial recommendations based on the prediction results.
- 5. To evaluate the model's performance using standard machine learning evaluation metrics.

METHODOLOGY

APPROACH AND METHODOLOGY TO BE USED

The project will follow the Cross-Industry Standard Process for Data Mining (CRISP-DM), a structured methodology widely used in machine learning projects. It provides a systematic approach to developing the prediction model, from understanding the problem to evaluating the final results.

Phase	Description
Business Understanding	Define the project's objectives, identify user needs, and determine how machine learning can help predict overspending risks.
Data Understanding	Collect and examine expense datasets to identify relevant features, trends, and potential data quality issues.
Data Preparation	Clean, preprocess, and transform the collected data before training the machine learning model.
Modeling	Train and optimize a Random Forest model using the prepared dataset.
Evaluation	Assess the model's performance using appropriate evaluation metrics to determine its prediction accuracy and reliability.
Deployment	Integrate the trained model into the web application to generate overspending predictions and financial recommendations.

JUSTIFICATION

- CRISP-DM provides a structured workflow for developing machine learning applications.

- Random Forest is effective for classification problems, handles large datasets efficiently, and reduces the risk of overfitting.
- A web-based application allows users to access the system through a browser without installing additional software.

DATA COLLECTION AND PREPARATION

SOURCES OF DATA

The machine learning model will be trained using publicly available datasets and simulated expense records.

Data Source	Purpose
Kaggle Personal Finance Datasets	Provides real-world expense records for training and testing the model.
Open Financial Datasets	Supplies additional financial data to improve model reliability and support feature analysis.
Simulated Expense Data	Supplements the dataset by representing common spending behaviors that may not be sufficiently covered by public datasets.

PREPROCESSING STEPS

Before training the machine learning model, the collected data will undergo the following preprocessing steps:

1. **Data Cleaning** – Remove duplicate, incomplete, and inconsistent records.
2. **Handling Missing Values** – Replace or remove missing values to improve data quality.

3. **Feature Encoding** – Convert categorical data, such as expense categories, into numerical values that can be processed by the model.
4. **Feature Scaling** – Normalize numerical values when necessary to improve model performance.
5. **Feature Selection** – Select relevant variables that contribute to overspending prediction.
6. **Dataset Splitting** – Divide the dataset into training, validation, and testing sets for model development and evaluation.

MODEL DEVELOPMENT

ALGORITHMS AND TOOLS TO BE USED

The proposed system uses the Random Forest algorithm to classify a user's overspending risk based on historical spending behavior. Random Forest was selected because it performs well in classification tasks, handles different types of input features, reduces overfitting through multiple decision trees, and provides reliable prediction results. These characteristics make it suitable for analyzing spending patterns and identifying users who may be at risk of overspending.

The following tools will be used throughout the development process:

Tool	Purpose
Python	Machine learning model development and backend programming
Scikit-learn	Training and evaluating the Random Forest model
Pandas	Data cleaning, preprocessing, and analysis
NumPy	Numerical computations and data manipulation
Flask	Backend framework for integrating the machine learning model with the web application
HTML, CSS, Bootstrap	Development of the web-based user interface
Visual Studio Code	Source code development and debugging
GitHub	Version control and collaboration

TRAINING AND VALIDATION PROCESS

The collected dataset will be divided into three subsets:

- **Training Set (70%)** – Used to train the Random Forest model by learning spending patterns from historical expense records.
- **Validation Set (15%)** – Used to evaluate the model during training and adjust parameters to improve prediction performance.
- **Testing Set (15%)** – Used to measure the model's performance on unseen data and evaluate its ability to generalize to new spending records.

During training, the model will learn relationships between features such as expense amount, category, transaction frequency, and spending history. Once trained, the model will generate predictions indicating whether a user's spending behavior suggests a low, moderate, or high risk of overspending.

SOFTWARE INTERFERENCE

TOOLS USED FOR PROTOTYPING

The user interface will be designed before system development to ensure an organized and user-friendly experience. The following tools will be used during development and prototyping.

Tool	Purpose
Figma	Wireframes, UI mockups, and interface prototyping
Visual Studio Code	Frontend and backend development
Flask	Backend development and machine learning integration
Python	Machine learning implementation
GitHub	Version control and team collaboration

DESKTOP / MOBILE APPLICATION UI SAMPLE MOCKUPS

1. Login Page

The Login page allows registered users to securely access the system using their account credentials. New users may also create an account before accessing the dashboard.

AI Spending Habit Analyzer

[LOGO]

Welcome Back!

Email

Password

☐ Remember Me

[Login]

Don't have an account?

[Register Here](#)

Figure 4. Login Page Mockup

2. Dashboard

The Dashboard provides an overview of the user's financial activity. It displays recent expenses, spending summaries, prediction results, and quick access to the system's main features.

LOGO Dashboard Profile

Total Expenses

₱12,450

Monthly Budget

₱15,000

Remaining Budget

₱2,550

Recent Expenses

Food ₱350

Transport ... ₱80

Shopping ... ₱950

[Pie Chart]

Food

Transport

Bills

Shopping

Overspending Risk

🟡 Moderate Risk

Recommendation:

Reduce discretionary spending by 15%.

Figure 5. Dashboard Mockup

3. Add Expense Page

The Add Expense page allows users to record new expense transactions by entering the amount, selecting a category, choosing the transaction date, and adding optional notes.



The image shows a mobile application form titled "Add Expense". The form is enclosed in a light gray container with dashed lines at the top and bottom. It contains the following fields and controls:

- Add Expense**: The title of the form.
- Amount**: A label for the input field.
- : A text input field for the amount, containing the letter "P".
- Category**: A label for the dropdown menu.
- ▼ Food**: A dropdown menu showing "Food" as the selected category.
- Date**: A label for the date input field.
- [07/11/2026]**: A date input field showing the date "07/11/2026".
- Notes**: A label for the text area.
- : A text input field for notes, currently empty.
- [Save Expense]**: A button to save the expense.

Figure 6. Add Expense Page

4. Prediction Results Page

The Prediction Results page displays the user's predicted overspending risk together with personalized financial recommendations based on the machine learning model's analysis.

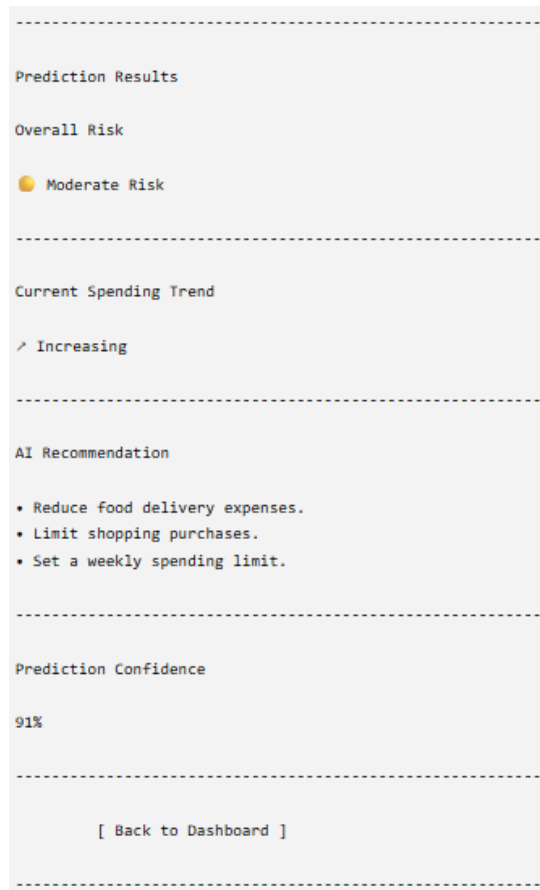


Figure 7 Prediction Results Page Mockup

3-TIER ARCHITECTURE DESIGN

The AI Spending Habit Analyzer follows a three-tier architecture consisting of the Presentation Layer, Application Layer, and Data Layer. This architecture separates the user interface, business logic, and data management, making the system easier to maintain, scale, and update.

Layer	Function
Presentation Layer	Provides the web interface where users manage expenses, view reports, and access prediction results.
Application Layer	Processes user requests, executes business logic, and communicates with the Random Forest model to generate predictions.
Data Layer	Stores user accounts, expense records, prediction history, and datasets used by the machine learning model.

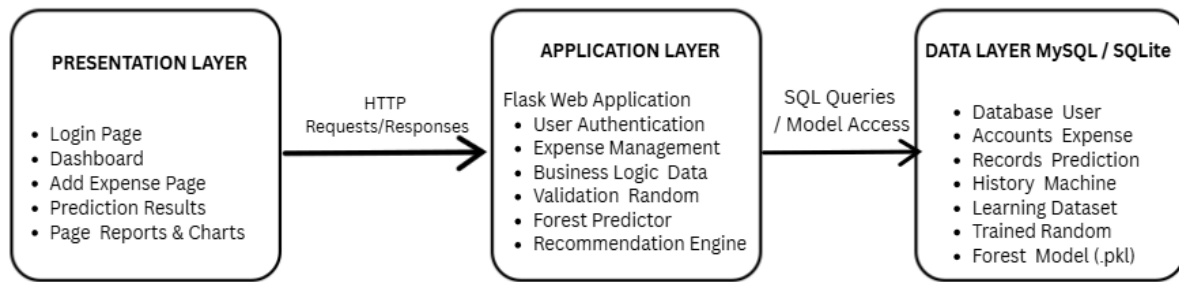


Figure 8. Three-Tier Architecture of the AI Spending Habit Analyzer.

MODEL PERFORMANCE EVALUATION

METRICS AND EVALUATION METHODS

The performance of the Random Forest model will be evaluated using standard classification metrics to determine its accuracy and reliability in predicting overspending risks. These metrics provide a comprehensive assessment of the model's ability to correctly classify users based on their spending behavior.

Metric	Purpose
Accuracy	Measures the overall percentage of correct predictions made by the model.
Precision	Measures how many users predicted to be at risk of overspending are actually at risk.
Recall	Measures the model's ability to correctly identify users who are at risk of overspending.
F1-Score	Balances Precision and Recall to evaluate overall classification performance.
Confusion Matrix	Summarizes correct and incorrect classifications to identify prediction errors.
ROC-AUC Score	Measures the model's ability to distinguish between different overspending risk levels.

The evaluation results will be used to compare the model's prediction performance and determine whether further improvements or parameter adjustments are necessary before deployment.

EXPECTED OUTCOMES

The proposed system is expected to provide an intelligent expense management platform that helps users understand and improve their spending habits through machine learning.

Upon completion, the project is expected to deliver:

- A web-based expense management system with secure user authentication.
- A machine learning model capable of predicting overspending risks using historical expense data.
- An interactive dashboard that displays expense summaries, spending trends, and prediction results.
- Personalized financial recommendations based on the user's spending behavior.
- A structured and maintainable system architecture that supports future enhancements.

The project also aims to promote better financial awareness by helping users recognize spending patterns, make informed budgeting decisions, and reduce the likelihood of overspending.

PROJECT TIMELINE

The project will be completed following the development schedule prepared by the project team. The timeline outlines the major activities from project planning to final documentation and ensures that each phase is completed within the allotted timeframe.

[Insert Figure 9. Project Schedule / Gantt Chart Snapshot]

GITHUB PROJECT REPOSITORY

Link: https://github.com/JacinthUlpiah/CPE178_E01_ProjectProposal_Group02_3T2526.git

APPENDICES

APPENDIX A. USE CASE DIAGRAM

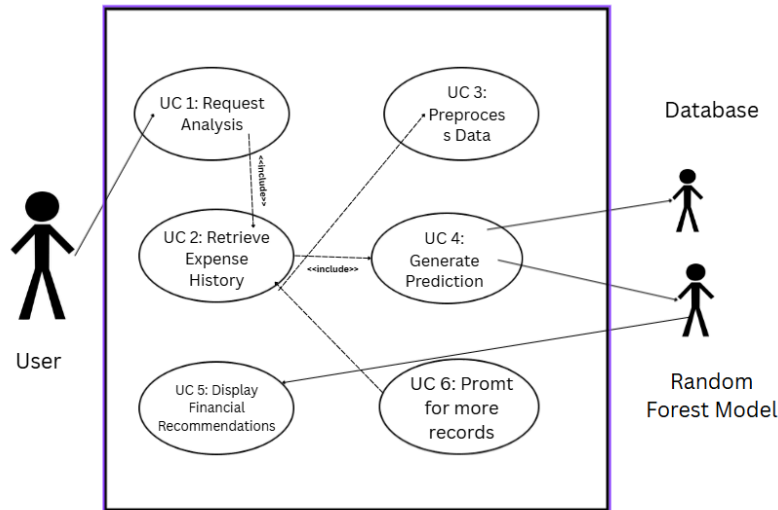


Figure A: Use Case Diagram of the AI Spending Habit Analyzer.

APPENDIX B. FULL USE CASE NARRATIVES

Use Case 1 - Record Expense

Field	Description
Use Case Name	Record Expense
Primary Actor	User
Description	Allows the user to record daily expenses by entering the transaction amount, category, date, and optional notes. The recorded expense is stored in the database and becomes available for reports and machine learning analysis.
Preconditions	The user has successfully logged into the system.
Trigger	The user selects Add Expense from the dashboard.
Main Flow	1. The user opens the Add Expense page.2. The system displays the expense entry form.3. The user enters the amount, category, date, and optional notes.4. The user submits the form.5. The system validates the input data.6. If valid, the system stores the expense in the database.7. The dashboard updates the expense summary and recent transactions.8. The system displays a confirmation message.

Alternative Flow	5a. If required fields are incomplete or contain invalid values, the system displays an error message and requests the user to correct the input before submission.
Postconditions	The expense record is successfully saved and becomes available for reports, dashboard summaries, and overspending prediction.

Use Case 2 - Predict Overspending Risk

Field	Description
Use Case Name	Predict Overspending Risk
Primary Actor	User
Description	Allows the system to analyze the user's historical spending behavior using a trained Random Forest model and estimate the user's overspending risk.
Preconditions	The user is logged into the system and has sufficient expense records for analysis.
Trigger	The user selects Predict Overspending from the dashboard.
Main Flow	1. The user requests a prediction.2. The system retrieves the user's historical expense records.3. The system preprocesses the retrieved data.4. The Random Forest model analyzes the spending history.5. The system generates the predicted overspending risk.6. Personalized financial recommendations are generated based on the prediction.7. The prediction result and recommendations are displayed on the dashboard.
Alternative Flow	3a. If insufficient expense records are available, the system informs the user that additional spending history is required before a prediction can be generated.
Postconditions	The overspending prediction and financial recommendations are successfully displayed to the user.

Appendix C. Class Diagram

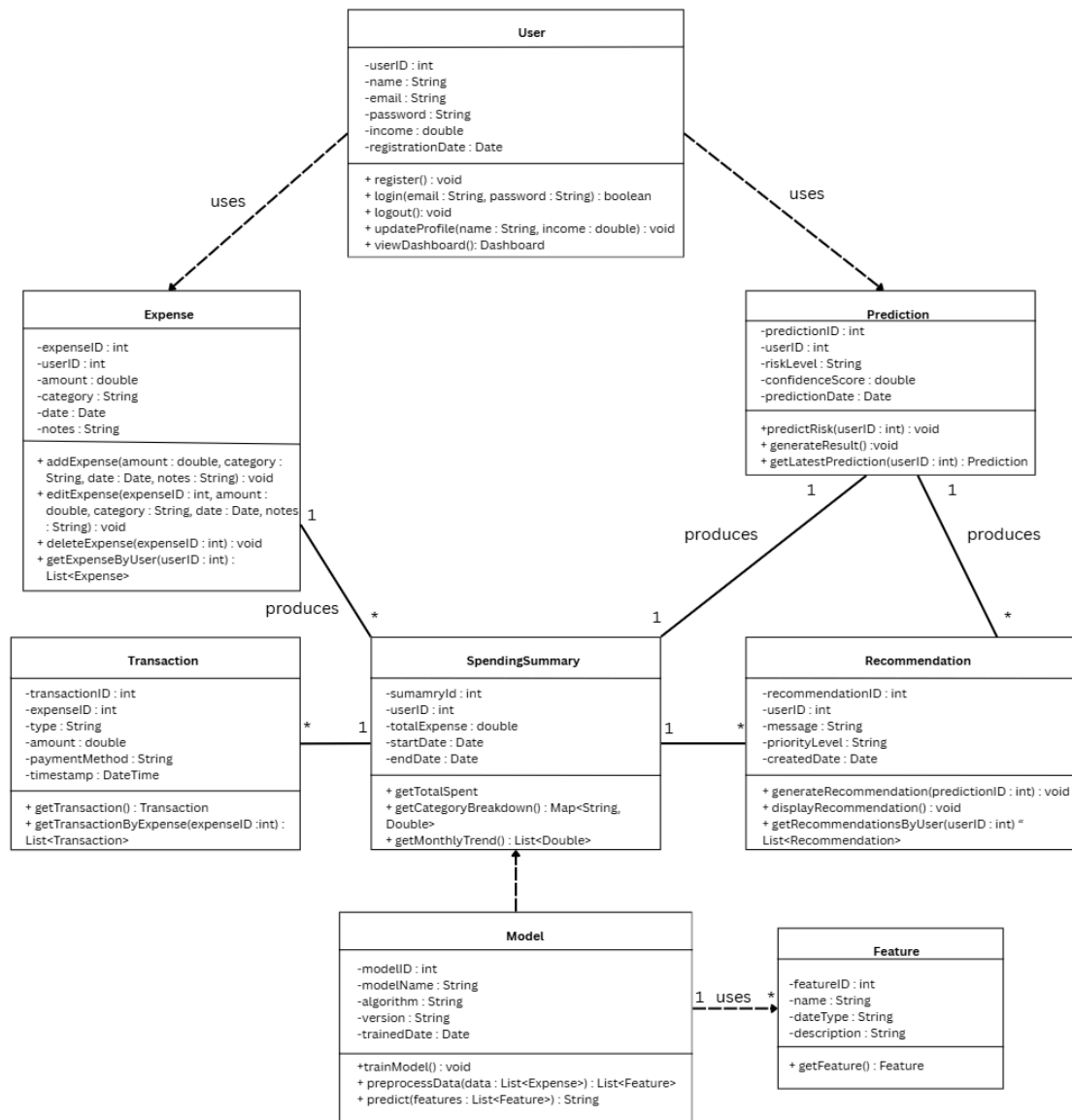


Figure C.1. Class Diagram of the AI Spending Habit Analyzer.

Appendix D. Activity Diagrams

Activity Diagram 1 – Record Expense



Figure D.1. Activity Diagram for the Record Expense use case.

Activity Diagram 2 - Predict Overspending Risk

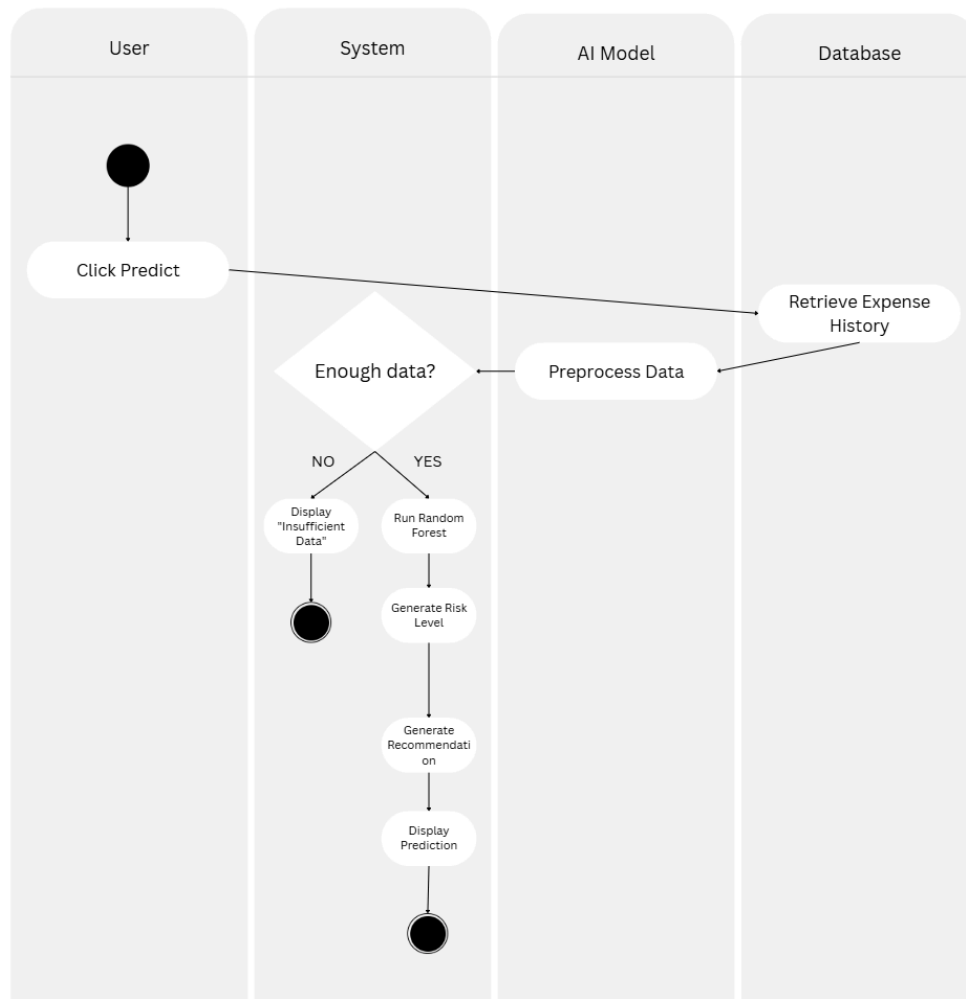


Figure D. Activity Diagram for the Record Expense use case.

Appendix E. Sequence Diagram

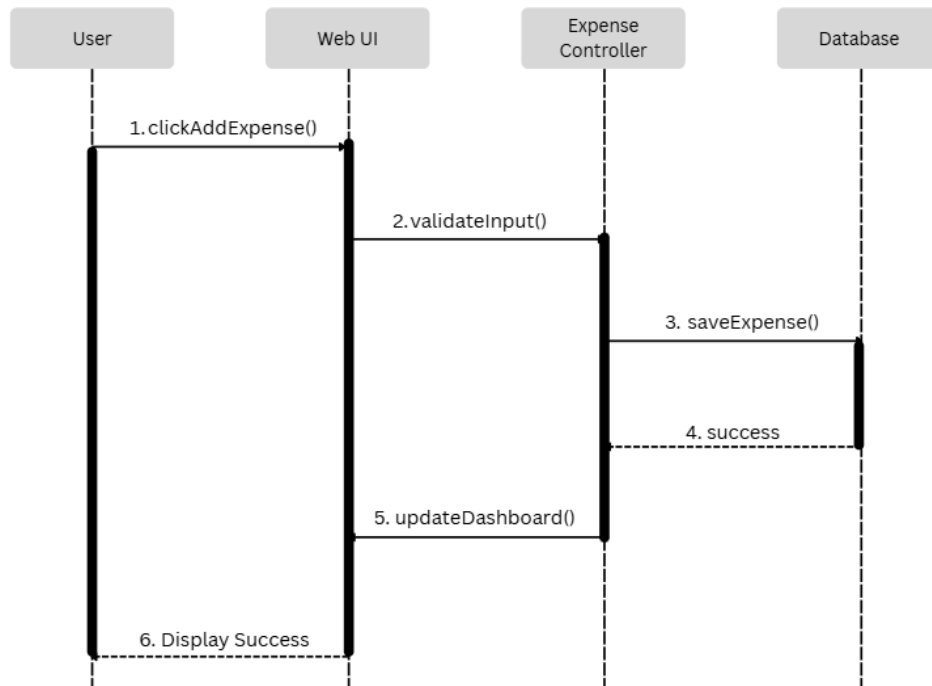


Figure E. Sequence Diagram for recording an expense

Appendix F. Traceability Matrix

Requirement ID	Requirement	Use Case	Test Case
FR-01	User Registration and Login	User Login	TC-01
FR-02	Record Expense	Record Expense	TC-02
FR-03	Edit Expense	Record Expense	TC-03
FR-04	Delete Expense	Record Expense	TC-04
FR-05	Predict Overspending Risk	Predict Overspending Risk	TC-05
FR-06	Generate Financial Recommendations	Predict Overspending Risk	TC-06
FR-07	Display Dashboard	Record Expense	TC-07
FR-08	View Spending Reports	Record Expense	TC-08

Requirements Traceability Matrix

Appendix G. Test Cases

Test Case ID	Scenario	Input	Expected Result	Pass/Fail Criteria
TC-01	User Login	Valid username and password	User successfully logs into the system	Dashboard is displayed.
TC-02	Add Expense	Valid expense details	Expense is saved successfully	Expense appears in the dashboard and history.
TC-03	Edit Expense	Updated expense details	Expense record is updated	Updated information is displayed correctly.
TC-04	Delete Expense	Existing expense record	Expense is removed	Deleted expense no longer appears in the records.
TC-05	Predict Overspending	Sufficient expense history	Prediction is generated	Risk level and recommendation are displayed.
TC-06	Insufficient Data	Limited expense history	Prediction is not generated	System displays an informative message.
TC-07	View Dashboard	User logs in	Dashboard loads successfully	Expense summary and charts are displayed correctly.
TC-08	View Reports	Existing expense records	Spending reports are generated	Reports display accurate expense data.

System Test Cases

Appendix H. Three-Tier Architecture Diagram

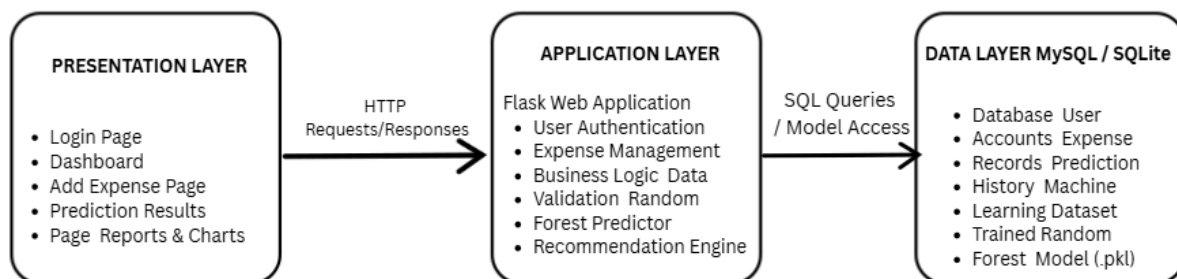


Figure H. Three-Tier Architecture of the AI Spending Habit Analyzer.

